Trainwreck

A subtle data poisoning attack

Key Features

- Poisoning the Training Data
- Adversarial Perturbations
- Model Misclassification
- Stealthy
- Black-Box
- Transferable

Steps to Perform Trainwreck Attack

- Dataset Poisoning
- Model Training on Poisoned Data
- Training and Validation
- Deployment
- Exploitation

Implications

- Cybersecurity
- Autonomous Driving
- Facial Recognition
- Healthcare

Demonstration

Due to how long it takes to run – I will not demo in class, however, I can later if anyone would like.

Required Libraries

cleverhans

scipy

timm

tqdm

torch (with CUDA)

torchvision

pip install -r requirements.txt

https://pytorch.org/get-started

Performing Trainwreck

Steps to perform the Trainwreck Attack (using CIFAR-10 dataset)

1. Feature Extraction – Optional
2. Training the Surrogate Model
3. Crafting the Attack
4. Executing the Attack
5. Results Analysis

```
python feature_extraction.py <DATASET_ID> --batch_size <BATCH_SIZE>
python attack_and_train.py clean <DATASET_ID> surrogate --batch_size <BATCH_SIZE> --
      n_epochs <N_EPOCHS>
python craft_attack.py <ATTACK_METHOD> <DATASET_ID> <POISON_RATE> --epsilon_px <EPS>
python attack_and_train.py <ATTACK_METHOD> <DATASET_ID> <TARGET_MODEL> --poison_rate
      <PI> --epsilon_px <EPS> --batch_size <BATCH_SIZE> --n_epochs <N_EPOCHS>
python results_analysis.py
```

- <DATASET_ID> is a valid dataset string ID recognized in datasets/dataset.py.
- ATTACK_METHOD> is the identifier of the attack method.
- <POISON_RATE>, or *π* in the paper, is the proportion of the training data to be poisoned.
- <TARGET_MODEL> is the model we are trying to attack by training it on the poisoned data.
- --epsilon_px is an optional parameter used by the perturbation attacks to denote the l-inf norm restriction on perturbation strength
- --batch_size is an optional parameter specifying the batch size the feature extractor model will use.
- --n_epochs is an optional parameter specifying the number of training epochs
- --force is an optional parameter that forces the script to execute even if a run with the same parameters has been completed before.
- --poison_rate, or *π* in the paper, is the proportion of the training data to be poisoned

Performing Trainwreck

Training the surrogate model

python attack_and_train.py clean cifar10 surrogate --batch_size 32 --n_epochs 30

```
[20 Oct 2024, 18:15:21] +++ TRAINWRECK STARTED +++
[20 Oct 2024, 18:15:21] Poisoning the data...
Files already downloaded and verified
Files already downloaded and verified
[20 Oct 2024, 18:15:23] Data poisoned in 1.53 seconds.
[20 Oct 2024, 18:15:23] +++ TRAINING cifar10-clean-surrogate-resnet50-30epochs +++
[20 Oct 2024, 18:16:14] Epoch 1/30 completed in 50.83 seconds: top-1 accuracy = 76.42, top-5 accuracy = 98.55, loss = 0.7. Best top-1 accuracy so far = 76.42.
[20 Oct 2024, 18:17:03] Epoch 2/30 completed in 48.82 seconds: top-1 accuracy = 76.0, top-5 accuracy = 97.22, loss = 0.78. Best top-1 accuracy so far = 76.42.
[20 Oct 2024, 18:17:52] Epoch 3/30 completed in 49.55 seconds: top-1 accuracy = 81.86, top-5 accuracy = 99.09, loss = 0.54. Best top-1 accuracy so far = 81.86.
[20 Oct 2024, 18:18:42] Epoch 4/30 completed in 49.49 seconds: top-1 accuracy = 80.27, top-5 accuracy = 98.95, loss = 0.72. Best top-1 accuracy so far = 81.86.
[20 Oct 2024, 18:19:32] Epoch 5/30 completed in 49.95 seconds: top-1 accuracy = 83.47, top-5 accuracy = 99.13, loss = 0.52. Best top-1 accuracy so far = 83.47.
[20 Oct 2024, 18:20:22] Epoch 6/30 completed in 49.76 seconds: top-1 accuracy = 82.84, top-5 accuracy = 99.1, loss = 0.54. Best top-1 accuracy so far = 83.47.
[20 Oct 2024, 18:21:12] Epoch 7/30 completed in 50.55 seconds: top-1 accuracy = 83.44, top-5 accuracy = 98.89, loss = 0.54. Best top-1 accuracy so far = 83.47.
[20 Oct 2024, 18:22:03] Epoch 8/30 completed in 50.77 seconds: top-1 accuracy = 83.11, top-5 accuracy = 99.14, loss = 0.56. Best top-1 accuracy so far = 83.47.
[20 Oct 2024, 18:22:54] Epoch 9/30 completed in 50.8 seconds: top-1 accuracy = 83.64, top-5 accuracy = 99.03, loss = 0.58. Best top-1 accuracy so far = 83.64.
[20 Oct 2024, 18:23:44] Epoch 10/30 completed in 50.35 seconds: top-1 accuracy = 83.86, top-5 accuracy = 99.12, loss = 0.54. Best top-1 accuracy so far = 83.86.
[20 Oct 2024, 18:24:36] Epoch 11/30 completed in 51.68 seconds: top-1 accuracy = 83.89, top-5 accuracy = 98.96, loss = 0.57. Best top-1 accuracy so far = 83.89.
[20 Oct 2024, 18:25:27] Epoch 12/30 completed in 51.33 seconds: top-1 accuracy = 84.04, top-5 accuracy = 99.1, loss = 0.62. Best top-1 accuracy so far = 84.04.
[20 Oct 2024, 18:26:19] Epoch 13/30 completed in 51.74 seconds: top-1 accuracy = 84.24, top-5 accuracy = 98.92, loss = 0.62. Best top-1 accuracy so far = 84.24.
[20 Oct 2024, 18:27:10] Epoch 14/30 completed in 51.46 seconds: top-1 accuracy = 81.56, top-5 accuracy = 98.59, loss = 0.7. Best top-1 accuracy so far = 84.24.
[20 Oct 2024, 18:28:02] Epoch 15/30 completed in 52.12 seconds: top-1 accuracy = 84.74, top-5 accuracy = 99.18, loss = 0.59. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:28:55] Epoch 16/30 completed in 52.4 seconds: top-1 accuracy = 82.9, top-5 accuracy = 98.76, loss = 0.68. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:29:47] Epoch 17/30 completed in 52.32 seconds: top-1 accuracy = 82.94, top-5 accuracy = 98.94, loss = 0.64. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:30:39] Epoch 18/30 completed in 51.69 seconds: top-1 accuracy = 84.32, top-5 accuracy = 98.85, loss = 0.64. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:31:30] Epoch 19/30 completed in 51.26 seconds: top-1 accuracy = 83.54, top-5 accuracy = 98.85, loss = 0.7. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:32:21] Epoch 20/30 completed in 51.17 seconds: top-1 accuracy = 83.97, top-5 accuracy = 99.07, loss = 0.68. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:33:12] Epoch 21/30 completed in 50.77 seconds: top-1 accuracy = 84.02, top-5 accuracy = 99.05, loss = 0.72. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:34:02] Epoch 22/30 completed in 49.98 seconds: top-1 accuracy = 83.32, top-5 accuracy = 98.8, loss = 0.7. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:34:53] Epoch 23/30 completed in 50.68 seconds: top-1 accuracy = 84.53, top-5 accuracy = 99.13, loss = 0.68. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:35:44] Epoch 24/30 completed in 50.98 seconds: top-1 accuracy = 83.85, top-5 accuracy = 98.92, loss = 0.74. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:36:34] Epoch 25/30 completed in 50.25 seconds: top-1 accuracy = 83.96, top-5 accuracy = 98.84, loss = 0.78. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:37:25] Epoch 26/30 completed in 50.67 seconds: top-1 accuracy = 77.7, top-5 accuracy = 97.93, loss = 0.91. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:38:15] Epoch 27/30 completed in 50.24 seconds: top-1 accuracy = 84.33, top-5 accuracy = 98.89, loss = 0.7. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:39:06] Epoch 28/30 completed in 51.22 seconds: top-1 accuracy = 83.9, top-5 accuracy = 99.0, loss = 0.77. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:39:57] Epoch 29/30 completed in 51.42 seconds: top-1 accuracy = 83.15, top-5 accuracy = 98.81, loss = 0.82. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:40:50] Epoch 30/30 completed in 52.68 seconds: top-1 accuracy = 84.39, top-5 accuracy = 98.81, loss = 0.75. Best top-1 accuracy so far = 84.74.
[20 Oct 2024, 18:40:50] +++ TRAINING COMPLETE +++
[20 Oct 2024, 18:40:50] +++ TRAINWRECK FINISHED (25 minutes, 29.04 seconds) +++
```

Defending against Trainwreck

- Data sanitization and verification
- Robust training techniques
- Model validation